CVPR 2025 · ORAL PRESENTATION

AIpparel

A Multimodal Foundation Model for Digital Garments

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CVPR 2025

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OUTLINE

Four acts across eighteen slides

I

ACT I

The problem

Why prior single-modal pattern generators stall on complex garments.

slides 3 – 5

II

ACT II

AIpparel — the model

A drawing-command tokenizer, a multimodal dataset, and a mixed-loss objective.

slides 6 – 11

III

ACT III

Does it work?

Image-to-pattern, multimodal prompts, natural-language editing, and one ablation.

slides 12 – 16

IV

ACT IV

Takeaways

What worked, what remains, and the release plan.

slides 17 – 18

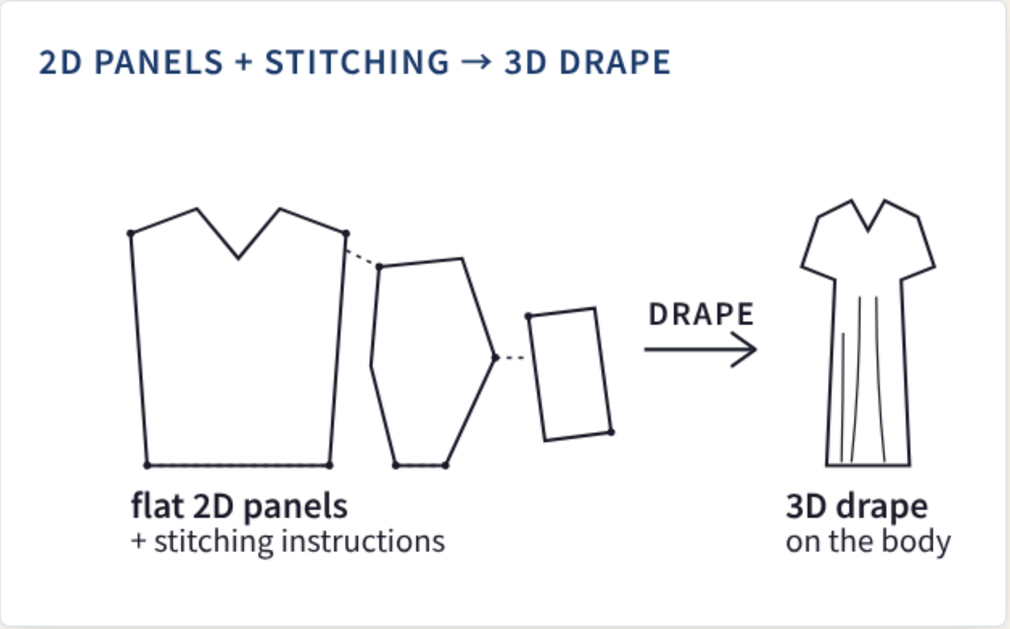
Digital garments need multimodal pattern-making

Sewing patterns bridge 2D design and 3D drape — but pattern-making is still manual and iterative.

> Even an experienced tailor *goes through multiple iterations*, folding in **verbal fit descriptions** and **visual references** before a pattern lays right on the body.

— PARAPHRASED FROM AIPPAREL § 2.2 • INTRODUCTION, P. 1

Clothing is a barrier against the elements, a reflection of societal norms, and a means of personal expression — yet the craft that turns intent into a wearable still runs on manual pattern-making.



LIMITATIONS OF PRIOR WORK

Three gaps stop prior methods short

Single modality • No multimodal dataset • Simple garments only

01

Locked to one input modality

Sewing-pattern predictors are engineered around either 3D points, or a single image, or a text prompt — and are hard to adapt when the task combines them.

MATERIAL §2.3 & §3.1

02

No dataset carries text + image + edits

DressCode, SewFactory and GarmentCodeData each annotate one or two modalities. No public corpus pairs all three, so multimodal supervision has no place to come from.

MATERIAL §3.3

03

Multimodal prior confined to simple garments

The one multimodal predecessor (Wang et al.) accepts image + edits but is bound to a hand-authored parameter set — expressive enough only for a narrow garment vocabulary.

MATERIAL §3.4

MODALITY AVAILABILITY

What each dataset actually annotates

Adapted from Table 1 · text · image · edits

DATASET	TEXT	IMAGE	EDITS
Wang et al.	✗	✓	✓
Korosteleva & Lee	✗	✓	✗
SewFactory	✗	✓	✗
DressCode	✓	✗	✗
GarmentCodeData	✗	✓	✗
GCD-MM (Ours)	✓	✓	✓

Only GCD-MM ticks all three — the empty column that motivates a new dataset.

FIGURE 1 · P. 1

One model, four use cases, simulation-ready patterns

Text • Image • Text + Image • Edit instruction — every path lands on a pattern that drapes in 3D.

01 Text	02 Image	03 Text + Image	04 Edit instruction	EXAMPLE PROMPT
				“A knee-length jumpsuit with long sleeves.”



Fig. 1, p. 1 · reproduced from Nakayama et al., AIpparel, CVPR 2025

OUR CONTRIBUTIONS

One dataset, one tokenizer, one multimodal model

The three moving parts of AIpparel — each with its quantitative anchor.

01 DATASET

GCD-MM — a multimodal sewing-pattern dataset

Extends the largest public sewing-pattern dataset with paired multimodal annotations.

120,000+

unique garments • text + image + edits + editing instructions

02 TOKENIZER & TRAINING

Drawing-command tokenizer + new training objective

A novel tokenization scheme and objective for fine-tuning LMMs to predict sewing patterns.

~100× fewer tokens

avg ≈ **250** (max 838) vs **> 30,000** for DressCode

03 MODEL

AIpparel — the first multimodal foundation model for sewing patterns

Accepts language, images, and sewing patterns as input to predict simulation-ready patterns.

LLaVA-1.5-7B

fine-tuned backbone • first LMM-based sewing-pattern model

ROADMAP

07 Dataset → 08 Tokenizer → 09–11 Model & loss

Material §2.4 · p. 2

Build on GCD, add captions, add editing pairs

120 000 garments • text + image + editing annotations • rule-based short phrases + GPT-4o + programmatic edits.

01 EXTEND

Start from GarmentCodeData (GCD)

130 k procedurally generated sewing patterns, each with rendered draped-garment views.

Base corpus • Korosteleva & Lee, GCD [30]

02 CAPTION

Rule-based short phrases → GPT-4o caption

A rule-based algorithm reads each pattern's design parameters into short phrases; GPT-4o composes them with the rendered views into the final natural-language description.

`midi dress` `godet skirt` `flared hem` `V-neckline`

03 EDIT PAIRS

GarmentCode programmatic edits

Paired sewing patterns produced by the GarmentCode programming abstraction, each paired with a text instruction describing the applied edit.

`skirt length` `pants length` `insert style` `neckline style` `hood ±` `sleeve ±`

Material §4.1–4.4 · Table 1, p. 3

TABLE 1 · P. 3

Only GCD-MM carries all three modalities

Text / Image / Edits — has that modality?

DATASET	PATTERNS	TEXT	IMAGE	EDITS
WANG ET AL.	8 k	✗	✓	✓
KOROSTELEVA & LEE	23.5 k	✗	✓	✗
SEWFACTORY	19.1 k	✗	✓	✗
DRESSCODE	20.3 k	✓	✗	✗
GCD	130 k	✗	✓	✗
GCD-MM OURS	120 k	✓	✓	✓

TOKENIZATION

Sewing patterns as drawing commands

~100× fewer tokens than prior schemes on GCD-MM.

GARMENT 2	**<SoG>** start of garment	**<EoG>** end of garment		
PANEL 3	**<SoP>** start of panel	**<EoP>** end of panel	**<R>** 3D transform slot	
EDGE 8 (c…) = closing edge	**<L>** line	**<cL>** line, closing	**<Q>** quadratic Bézier	**<cQ>** quadratic, closing
	<B> cubic Bézier	**<cB>** cubic, closing	**<A>** arc	**<cA>** arc, closing
STITCHING M+1	**<t1> … <tM>** edges sharing a tag are stitched	**<tN>** null stitching tag		

Example panel — two stitched lines, one cubic Bézier, one arc

<SoP> [panel name] **<R>** <L> **<t1>** <L> **<t1>** <B> **<tN>** <A> **<tN>** **<EoP>**

Material §5.3 · Section 3.2, p. 4

TOKENS PER SEWING PATTERN ON GCD-MM

AIpparel

250 AVG

max **838**

DressCode / He et al.

> 30 000

fixed-length, zero-padded

COMPRESSION RATIO

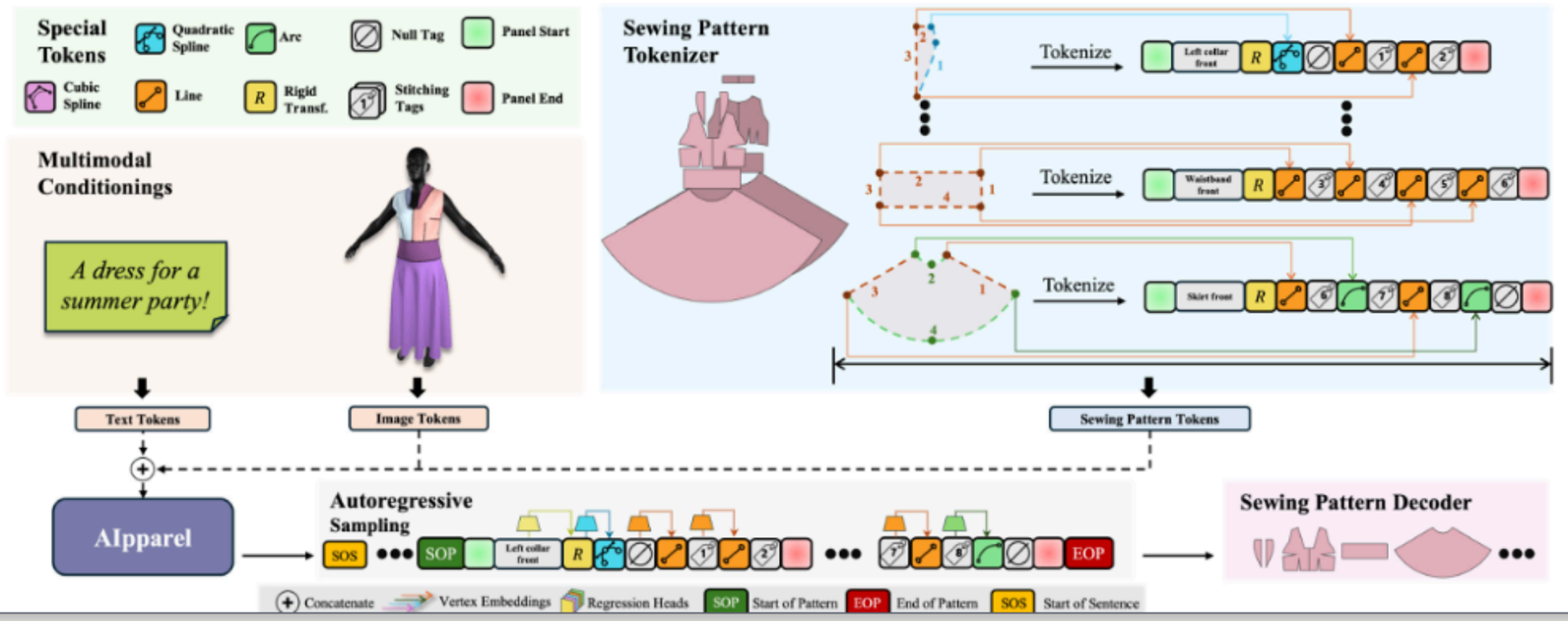
~100×

30 000 ÷ 250 ≈ 120 — the paper reports it as **around 100 times fewer**.

Unlocks LMM fine-tuning: pattern fits inside a 7B model's context.

End-to-end multimodal architecture

Multimodal input → Sewing-pattern tokenizer → LLaVA-1.5-7B → Autoregressive sampling → Simulation-ready pattern.



Reproduced from Nakayama et al., AIpparel, CVPR 2025 · Fig. 2, p. 4.

01 Special-token legend

02 Multimodal input (text + image)

03 Sewing-pattern tokenizer

04 AIpparel LMM (LLaVA-1.5-7B)

05 Autoregressive sampling → decoder

METHOD · CONTINUOUS PARAMETERS

Regression heads + positional embeddings

Small MLPs inject and predict the continuous fields the discrete tokens cannot carry.

A · REGRESSION HEAD

Edge regression head

$g_\theta^{(e)} : \mathbb{R}^D \to \mathbb{R}^C$

Reads the state **before** an edge token; predicts v_2 and control points.

B · REGRESSION HEAD

Transform regression head

$g_{\theta'}^{(R)} : \mathbb{R}^D \to \mathbb{R}^7$

Before a `<R>` token, predicts a panel's 3D transform: translation T + rotation q.

C · POSITIONAL EMBEDDINGS

Continuous → embedding

$h_\phi^{(e)} : \mathbb{R}^2 \to \mathbb{R}^D$

$h_{\phi'}^{(R)} : \mathbb{R}^7 \to \mathbb{R}^D$

Projects v_2 and (t, q) to dim D; **adds** to the token embedding.

Frozen vs trained

FINE-TUNING SCOPE · MATERIAL §5.6 · SECTION 3.2, P. 5–6

FROZEN · 2

- ✗ Vision encoder
- ✗ Vision → language projection

TRAINED · 4

- ✓ **Language transformer** f_ϕ
- ✓ **Edge regression head** $g_\theta^{(e)}$
- ✓ **Transform regression head** $g_{\theta'}^{(R)}$
- ✓ **Positional-embedding MLPs** $h_\phi^{(e)}$, $h_{\phi'}^{(R)}$

TRAINING OBJECTIVE

One backbone, one checkpoint, one mixed loss

Cross-entropy on discrete tokens + L2 on continuous edge / transform regressions.

FINE-TUNING LOSS

Eq. 2 · Material §5.7

CE

$$L_{CE}(f_\phi(X),\ X_{-1})$$

Cross-entropy on next-token prediction over the discrete drawing-command vocabulary.

+

L2 EDGE

$$\Sigma_{e'}\ \lambda_{e'}\ \cdot\ \|\ g_\theta^{(e)}(H_{<i})\ -\ v^2_{e'}\ \|^2$$

L2 on regressed edge endpoints and control points, summed over all edges.

+

L2 TRANSFORM

$$\Sigma_{R'}\ \lambda_{R'}\ \cdot\ \|\ g_{\theta'}^{(R)}(H_{<i})\ -\ R'\ \|^2$$

L2 on regressed per-panel transformation (translation + quaternion), summed over all panels.

λ_e, λ_R — numeric weights deferred to supplementary.

Material §5.7 & §6 · Section 3.2–4, p. 6

TRAINING SET-UP

BACKBONE
LLaVA-1.5-7B

VISION ENCODER + PROJECTION
Frozen

TRAIN : VAL : TEST
90 : 5 : 5

CHECKPOINT
Single unified — image, text, editing

OPTIMIZER • LR • BATCH • EPOCHS • TRAIN GPUS • WALL-CLOCK
Deferred to supplementary

INFERENCE PROFILING
Single NVIDIA A6000 — Tab. 5, inference only

EVALUATION

Four tasks, three baselines families, one unified checkpoint

Same AIpparel checkpoint evaluated on image-to-pattern, multimodal, editing, and a tokenizer ablation.

TASK	DATASET(S)	BASELINES	METRICS
01 **Image-to-Pattern**	SewFactory GCD-MM	SewFormer SewFormer-FT	Panel L2 ↓ cm #Panel Acc ↑ #Edge Acc ↑ Rot L2 ↓ Transl L2 ↓ cm #Stitch Acc ↑
02 **Multimodal Prediction**	100-sample benchmark	SewFormer-FT† DressCode†	Accuracy ↑ Panel L2 ↓ cm
03 **Language-driven Editing**	GCD-MM editing set	Sewformer* DressCode*	Accuracy ↑ Panel L2 ↓ cm
04 **Tokenizer Ablation**	DressCode test set	DressCode Ours w.o. reg.	Accuracy ↑ Panel L2 ↓ cm Time ↓ s · A6000

↑ higher is better ↓ lower is better ● One AIpparel checkpoint spans tasks 1–3; task 4 tests the tokenizer + regression design. MATERIAL §7 · P. 6–8

FT fine-tuned on GCD-MM • † LMM adapters (DALL • E, GPT-4o) • * Edit adapters (InstructPix2Pix, GPT-4o) • **Accuracy** = #Panel Acc × #Edge Acc.

State-of-the-art on both datasets — decisive on the harder one

GCD-MM #Stitch Accuracy: 77.2 % vs 2.8 % — a 27× gap on the metric that matters for simulation.

METHOD	PANEL L2 ↓ cm	#PANEL ACC ↑ %	#EDGE ACC ↑ %	ROT L2 ↓ quat.	TRANSL L2 ↓ cm	#STITCH ACC ↑ %
SEWFACTORY — SewFormer's home distribution						
SEWFORMER	3.3	89.8	99.3	.008	0.8	99.2
AIPPAREL OURS	2.8	93.9	99.9	.005	0.6	99.8
GCD-MM — harder garments, diverse edge topology						
SEWFORMER-FT	12.3	79.4	44.7	.040	4.5	2.8
AIPPAREL OURS	5.4	85.2	82.7	.020	2.7	77.2

Table 2, p. 5 · Material §8.1

AIpparel (Ours) · Winning cell per metric · ↓ lower is better · ↑ higher is better

GCD-MM • #STITCH ACCURACY

77.2% AIpparel vs **2.8%** SewFormer-FT

27× gap on the metric that decides whether a pattern is simulation-ready

`Stitch Accuracy` correctness of predicted stitching pairs vs ground truth (§ 4). SewFormer-FT's fixed-length tokenizer cannot encode GCD-MM's small panels and diverse edge types — so its stitching collapses despite fine-tuning.

Adapters aren't enough — end-to-end multimodal wins

100-sample benchmark, 5 input categories × 20 samples each.

TABLE 3 • MULTIMODAL SEWING PATTERN PREDICTION

Single-modal baselines are wrapped with LMMs (GPT-4o, DALL • E) that translate multimodal input into the baseline's native modality before feeding.

METHOD	ACCURACY ↑ (%)	PANEL L2 ↓ (cm)
SEWFORMER-FT †	10.3	22.4
DRESSCODE †	0.6	31.0
AIPPAREL	59.0	6.1

AIpparel outperforms both LMM-augmented baselines by a large margin.

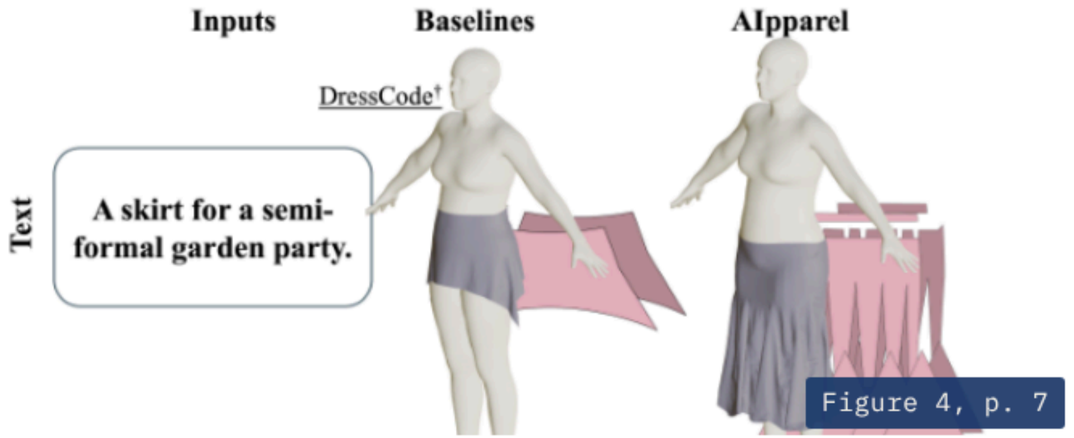


Figure 4, p. 7

REASONING PROMPT "semi-formal garden party"

DressCode † → generic mini-skirt

AIpparel → godet skirt matching the occasion

Precise edits without collateral drift

Two rows, two instructions, one style-preserving edit each.

PROMPTS `include a hood`

`make the skirt longer`

TABLE 4 • SEWING PATTERN EDITING

Method	Accuracy ↑ (%)	Panel L2 ↓ (cm)
Sewformer*	9.5	18.6
DressCode*	37.0	13.7
AIpparel	**83.4**	**1.5**

Sewformer wraps InstructPix2Pix around the input image before re-running SewFormer; DressCode* uses GPT-4o to rewrite the input caption.

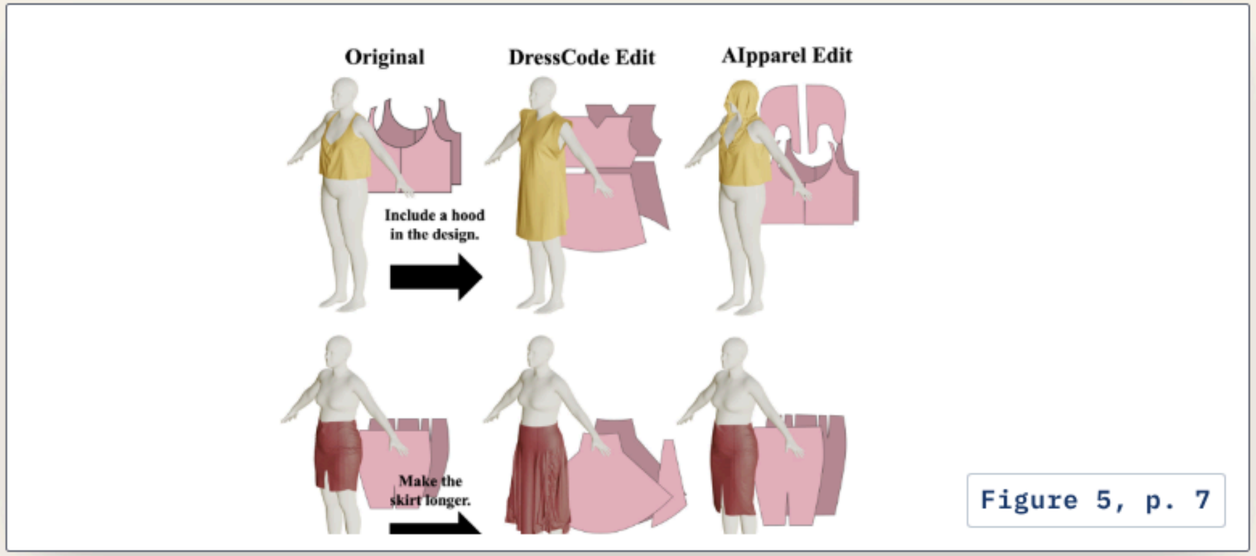


Figure 5, p. 7

"Our model follows the editing instructions more accurately compared with the baseline by accurately including a hood to the tank top (top row) and elongating the skirt (bottom row)."

DressCode restructures the tank top into a full-length dress and turns the tight skirt into a flared one.

ABLATION · TOKENIZER + LOSS

Tokenizer and regression loss carry the gain

DressCode test set (text-to-garment, 100 samples) • > 25× faster on a single A6000.

DressCode Fixed-length tokenizer • cross-entropy only.

Ours w.o. reg. Our tokenizer • regression heads removed, fields quantized into 256 bins.

Ours Full model • drawing-command tokenizer + regression heads (CE + L2).

Method	Accuracy ↑ (%)	Panel L2 ↓ (cm)	Time ↓ (s / garment)
DressCode	38.4	22.4	52.2
Ours w.o. reg.	79.0	7.2	3.4
Ours	**85.0**	**6.1**	**2.1**

52.2 s / 2.1 s ≈ 25× — paper caption: “more than 25× faster”.

Table 5 & Figure 6, p. 8 · Material §8.4 & §9.1

PROMPT “flared hem”

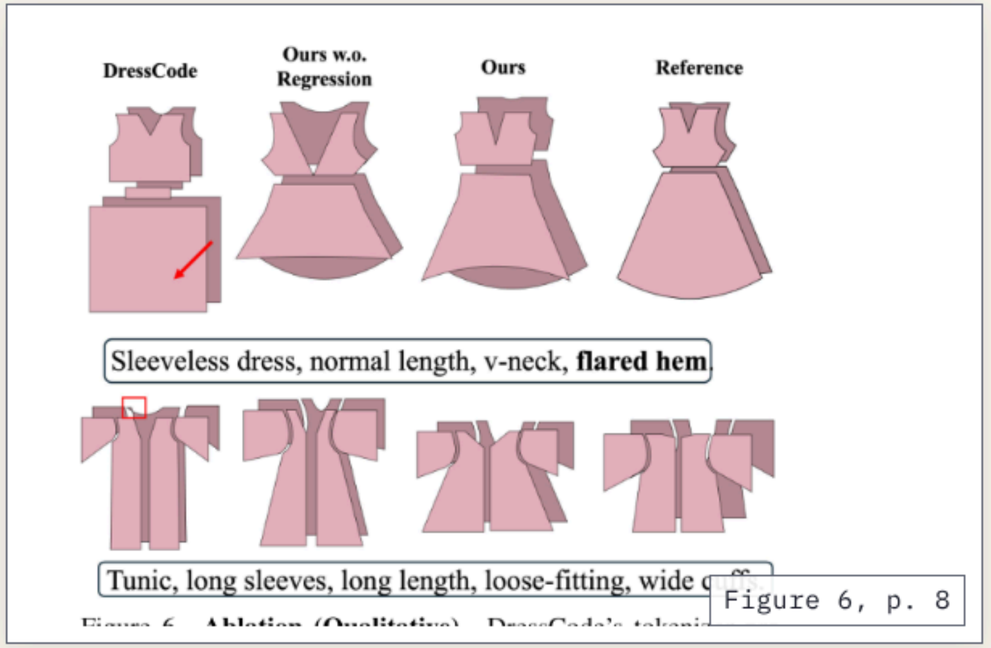


Figure 6, p. 8

DressCode's tokenizer produces unrealistic patterns **(red boxed region)** and does not match “flared hem”. Our tokenizer outputs geometrically regular panels that align with the prompt.

What worked • What remains

Three transportable ideas from the results, plus two limitations the paper flags itself.

TAKEAWAYS What worked

T-01 **Efficient tokenization is the unlock for scaling LMMs to complex geometries.**
Avg **~250 tokens/pattern** vs DressCode's **> 30,000** — **~100×** fewer, **> 25×** faster inference.

T-02 **A mixed CE + L2 objective makes continuous-parameter LMMs feasible.**
Regression heads on top of the tokenizer swap add **+6 pts** of pattern accuracy (Table 5).

T-03 **Multimodal training enables emergent common-sense reasoning.**
Text-to-pattern **59.0 %** vs **10.3 %** / **0.6 %** baselines; one checkpoint spans image, text, and edit.

LIMITATIONS What remains

L-01 **Representation is confined to manifold surfaces.**
> "The current representation… is still constrained to garments representable by manifold surfaces. Design elements like pockets require non-manifold structures."

L-02 **Physical fabrication is out of scope.**
> "Fabricating the generated garments is another interesting direction, which requires consideration of physical and material constraints during sewing pattern prediction."

Optimizer, learning rate, batch size, and training-time hardware are deferred to the paper's supplementary.

Material §5.3, §8.2, §8.4, §10.2 · Sections 3.2, 4.2, 4.4, 5, p. 4–8

THANK YOU

AIpparel — one model for text, image, and editing sewing-pattern generation

GCD-MM will be released publicly • future work: non-manifold structures + physical fabrication.

PUBLIC RELEASE

GCD-MM — the large-scale multimodal sewing-pattern dataset — will be released to enable further research in multimodal garment generation.

FUTURE WORK

- **Non-manifold** representations, e.g. pockets and multi-layer regions.
- **Physically-aware** fabrication — materials and physical constraints during prediction.